

DEEPAK KUMAR

Birla Institute of Technology, Mesra, Ranchi

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SUMMARY

Software Engineer with strong foundations in **Data Structures & Algorithms, Object-Oriented Programming, and System Design**. Experienced in developing scalable applications, automation systems, and performance-optimized software using **C++, Python, JavaScript**, and modern development frameworks. Proven ability to solve complex engineering problems through efficient algorithmic and software solutions.

EDUCATION

Birla Institute of Technology, Mesra

May 2027 (Expected)

Integrated M.Sc. in Mathematics and Computing | CGPA: 8.71/10

Ranchi, Jharkhand

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks.

TECHNICAL SKILLS

Languages: C++, Java, JavaScript (ES6+), Python, SQL

Backend & AI: Node.js, Express.js, FastAPI, LangChain, LangGraph, ChromaDB, MySQL, MongoDB

Frontend & Libraries: React.js, Tailwind CSS, Streamlit, Pydantic, PyPDF, Cheerio, Redux

DevOps & Tools: Git, GitHub Actions, Docker, Postman, Uvicorn, Python-dotenv, JWT

EXPERIENCE

Full Stack Developer (Freelance)

Oct 2023 – Dec 2023

LooksLove Salon

Remote | [Live]

- Developed and deployed a production-grade appointment management system with secure OTP-based authentication, serving **500+ active users**.
- Designed RESTful APIs and optimized database operations for appointment scheduling, improving system reliability and user experience.
- Built an administrative platform for inventory and staff management, streamlining operational workflows and reducing manual effort by **30%**.

PROJECTS

LectureLens AI | *Python, OpenCV, NumPy, Tkinter*

[Live]

- Designed and implemented a custom computer vision pipeline using Progressive Background Building to remove **95%** of visual obstructions from lecture recordings while preserving slide content.
- Improved processing efficiency by **40%** through a Dual-Resolution architecture that performed motion detection on low-resolution grayscale frames while rendering final outputs in full HD quality.
- Developed a desktop application capable of automatically generating high-quality PDF lecture notes from processed video content.

TnP Notifier | *Node.js, GitHub Actions, Cheerio, MongoDB*

[GitHub Link]

- Engineered an automated notification platform that continuously monitored institutional placement updates and delivered alerts to **200+ students**.
- Designed resilient GitHub Actions workflows with automated recovery mechanisms, maintaining **99.9%** service availability.
- Reduced manual monitoring effort by approximately **15 collective hours per week** through automated web scraping and event-driven notifications.

GenAI Chatbot Portfolio | *LangChain, FastAPI, ChromaDB, React.js*

[Live]

- Developed a document-query platform enabling semantic search and conversational interaction over resume datasets.
- Designed backend services using FastAPI and Pydantic for document ingestion, validation, vector indexing, and query processing.
- Implemented efficient retrieval pipelines and API integrations to support context-aware responses with low-latency performance.

PUBLICATIONS

OAuth 2.0 Security Analysis | *Paper ID: 328 (Under Review)*

- Authored a research paper analyzing authorization code flow vulnerabilities; proposed a token integrity protocol to mitigate **CSRF** and **Replay Attacks**.

ACHIEVEMENTS

- Solved **600+ DSA problems** across LeetCode and GeeksforGeeks.
- Secured the **Knight Badge** on LeetCode.
- Presented research on OAuth 2.0 security in healthcare systems at an IEEE conference hosted by NIT Jalandhar. [Certificate]
- Founded and operated **Earthen Pots**, managing product development, operations, and customer delivery.
- NCC 'C' Certificate holder demonstrating leadership, teamwork, and discipline.